

1. 题 $P_1 = P_2$ 2019 概率论 I 参考答案 (A)

(1) Proof. (a) 设 $z_n \xrightarrow{a.s.} z \iff 0 = \lim_{n \rightarrow \infty} P\left(\bigcup_{k=n}^{\infty} \{\omega: |z_k - z| \geq \varepsilon\}\right), \forall \varepsilon > 0$.

$$\Rightarrow P(\omega: |z_n(\omega) - z(\omega)| \geq \varepsilon) \leq P\left(\bigcup_{k=n}^{\infty} \{\omega: |z_k(\omega) - z(\omega)| \geq \varepsilon\}\right) \rightarrow 0$$

$$\text{i.e., } \lim_{n \rightarrow \infty} P(\{\omega: |z_n(\omega) - z(\omega)| \geq \varepsilon\}) = 0 \Rightarrow z_n \xrightarrow{P} z.$$

(b) 设 $z_n \xrightarrow{P} z$ holds.



$$\forall \underline{x} \in C(F_z(\cdot)) \text{ and } \forall y < x \Rightarrow$$

$$\{\omega: z < y\} = \{\omega: z < y, z_n < x\} \cup \{\omega: z < y, z_n \geq x\}$$

$$\subset \{\omega: z_n < x\} \cup \{\omega: |z_n - z| \geq x - y\}$$

$$\text{Then } F_z(y) \leq F_{z_n}(x) + P\{\omega: |z_n - z| \geq x - y\}$$

$$\text{Similarly, } F_{z_n}(x) \leq F_z(z) + P\{\omega: |z_n - z| \geq z - x\}, \forall z > x.$$

$$\text{Thus, } \left. \begin{aligned} F_z(y) &\leq \liminf_n F_{z_n}(x), \quad \forall y < x \\ \limsup_n F_{z_n}(x) &\leq F_z(z), \quad \forall x < z \end{aligned} \right\} \Rightarrow$$

$$F_z(y) \leq \liminf_n F_{z_n}(x) \leq \limsup_n F_{z_n}(x) \leq F_z(z), \quad \forall y < x < z$$

$$\text{取 } y \uparrow x \text{ and } z \downarrow x, \Rightarrow F_z(x-) = F_z(x) \leq \liminf_n F_{z_n}(x) \leq \limsup_n F_{z_n}(x) \\ \leq F_z(x+) = F_z(x)$$

$$\text{We have. } F_z(x) = \lim_n F_{z_n}(x), \quad \forall x \in C(F_z(\cdot)).$$

$$\text{i.e., } z_n \xrightarrow{W} z.$$

(2) Proof. (a) $Z_n \xrightarrow{P} Z \Leftrightarrow Z_n \xrightarrow{W} Z$ 不成立. 反例如下:

令 $\{Z, Z_n, n \geq 1\}$ 是 iid 随机列且 $Z \sim \begin{pmatrix} -1 & 1 \\ \frac{1}{2} & \frac{1}{2} \end{pmatrix}$

$$\text{则 } \forall \varepsilon > 0, P(\{|Z_n - Z| > \varepsilon\}) = P\{Z=1, Z_n=-1\} + P\{Z=-1, Z_n=1\}$$

$$= \frac{1}{4} + \frac{1}{4} = \frac{1}{2} > 0$$

$$\text{i.e., } Z_n \not\xrightarrow{P} Z$$

$$\text{但 } F_{Z_n}(x) = F_Z(x), \forall x \in \mathbb{R}, \Rightarrow \lim_n F_{Z_n}(x) = F_Z(x), \forall x \in C(F_Z)$$

$$\text{i.e., } Z_n \xrightarrow{W} Z.$$

(b) $Z_n \xrightarrow{a.s.} Z \Leftrightarrow Z_n \xrightarrow{P} Z$ 不成立. 反例如下:

$$\text{令 } \eta_n(\omega) = 1, \forall \omega \in \Omega = [0, 1), \mathcal{F} = \mathcal{B}(\Omega), P(d\omega) = dx.$$

$$\eta_{ki}(\omega) = \begin{cases} 1, & \omega \in [\frac{i-1}{k}, \frac{i}{k}), \quad i=1, 2, \dots, k. \\ 0, & \text{otherwise} \end{cases}$$

$$\text{Define } Z_1 = \eta_{11}, Z_2 = \eta_{21}, Z_3 = \eta_{22}, Z_4 = \eta_{31}, \dots$$

$$\text{则 } Z_n \not\xrightarrow{a.s.} 0 = Z, \because \exists Z_{k_1} = 1, Z_{k_2} = 0 \text{ for large}$$

$$k_1 \text{ and } k_2 \Rightarrow \lim_{k_1 \rightarrow \infty} Z_{k_1} = 1 \text{ and } \lim_{k_2 \rightarrow \infty} Z_{k_2} = 0$$

$$\text{but, } \lim_n P\{|Z_n| > \varepsilon\} = \lim_n P\left\{\left[\frac{i-1}{n}, \frac{i}{n}\right]\right\} = \lim_n \frac{1}{n} = 0.$$

$$\text{i.e., } Z_n \xrightarrow{P} 0 = Z.$$

$$\text{i.e., } Z_n \xrightarrow{P} Z \not\Rightarrow Z_n \xrightarrow{a.s.} Z.$$

"⇒"

(3) Proof. Assume that $f_n \xrightarrow{w} f$. Let $\varphi(x) = e^{itx}$. 则由 Helly 第 2 定理

和积分变换定理,

$$f_{f_n}(t) = E\{e^{itf_n}\} = \int_{\mathbb{R}} e^{itx} dF_{f_n}(x) = \int_{\mathbb{R}} e^{itx} dF_{f_n}(x) \rightarrow \int_{\mathbb{R}} e^{itx} dF_f(x)$$

$$= \int_{\mathbb{R}} e^{itx} dF_f(x) = f_f(t), \quad \forall t \in \mathbb{R}.$$

"⇐" 令 $F_n(\cdot) \triangleq F_{f_n}(\cdot)$, $F(\cdot) \triangleq F_f(\cdot)$. 由 Helly 第一定理, 序列

step one

$\{F_n(\cdot)\}$ 存在子列 $\{F_{n_k}(\cdot)\}$ st $F_{n_k} \xrightarrow{w} \hat{F}$, $\hat{F}$ 是非降左连续有界函数,

$\hat{F}(-\infty) \geq 0$, $\hat{F}(+\infty) \leq 1$. F 证 $\hat{F}(-\infty) = 0$ and $\hat{F}(+\infty) = 1$. 否则有

$$a \triangleq \hat{F}(+\infty) - \hat{F}(-\infty) < 1$$

由 $f(-\infty) = 0$ and $f(+\infty)$ 关于 f 一致连续, $\forall \varepsilon \in (0, 1-a)$, $\exists \tau(\varepsilon) > 0$

$$\text{st } \frac{1}{2\varepsilon} \left| \int_{-\tau}^{\tau} f(t) dt \right| > 1 - \frac{\varepsilon}{2} > a + \frac{\varepsilon}{2}$$

$$\text{and } h\varepsilon > \frac{4}{\varepsilon}$$

又 $F_{n_k} \xrightarrow{w} \hat{F} \Rightarrow \exists \pm b \in C(F(\cdot))$ and $k_0 \geq 1$ st $k > k_0$.

$$(\because a = \hat{F}(+\infty) - \hat{F}(-\infty) = \lim_k (F_{n_k}(b) - F_{n_k}(-b)))$$

$$F_{n_k}(b) - F_{n_k}(-b) < a + \frac{\varepsilon}{4}$$

$$\Rightarrow \frac{1}{2\varepsilon} \left| \int_{-\tau}^{\tau} f_{f_{n_k}}(t) dt \right| = \frac{1}{2\varepsilon} \left| \int_{-\infty}^{\infty} \left[\int_{-\tau}^{\tau} e^{itx} dt \right] dF_{f_{n_k}}(x) \right|$$

$$\leq \frac{1}{2\varepsilon} \int_{|x| < b} \left| \int_{-\tau}^{\tau} e^{itx} dt \right| dF_{f_{n_k}}(x) + \frac{1}{2\varepsilon} \int_{|x| > b} \left| \int_{-\tau}^{\tau} e^{itx} dt \right| dF_{f_{n_k}}(x)$$

$$\leq \int_{|x| < b} dF_{f_{n_k}}(x) + \frac{1}{2\varepsilon} \int_{|x| > b} \left| \frac{2}{x} \sin tx \right| dF_{f_{n_k}}(x)$$

$$\leq \int_{|x| < b} dF_{f_{n_k}}(x) + \frac{1}{2\varepsilon} \frac{2}{b} \leq F_{f_{n_k}}(b) - F_{f_{n_k}}(-b) + \frac{1}{h\varepsilon} \leq a + \frac{\varepsilon}{4} + \frac{\varepsilon}{4} = a + \frac{\varepsilon}{2}$$

又: $f_{3n_k}(t) \rightarrow f_3(t)$ and $|f_3(t)| \leq 1$, 由控制收敛定理, 令 $k \rightarrow \infty$

$$\frac{1}{2\pi} \left| \int_{-2}^2 f_3(t) dt \right| \leq a + \frac{\varepsilon}{2}$$

$$\text{与 } \frac{1}{2\pi} \left| \int_{-2}^2 f_3(t) dt \right| > a + \frac{\varepsilon}{2} \text{ 矛盾.}$$

$\therefore a=1$, i.e., $\hat{F}(-\infty)=0$ and $\hat{F}(+\infty)=1$ 同时成立.

i.e., $\hat{F}(\cdot)$ 为概率分布函数. 由Helly第2定理: $f_{3n_k}(t) \stackrel{n}{=} \int_{\mathbb{R}} e^{itx} d\hat{F}_{3n_k}(x)$

$$\rightarrow \int_{\mathbb{R}} e^{itx} d\hat{F}(x). \quad \text{又 } f_{3n_k}(t) \rightarrow f_3(t) \Rightarrow f_{3n_k}(t) \rightarrow f_3(t) = \int_{\mathbb{R}} e^{itx} d\hat{F}_3(x)$$

$$\Rightarrow f_3(t) = \int_{\mathbb{R}} e^{itx} d\hat{F}(x) = \int_{\mathbb{R}} e^{itx} d\hat{F}_3(x). \quad \text{由特征函数唯一性} \Rightarrow$$

$$\hat{F} \equiv \hat{F}_3. \quad \text{i.e., } F_{3n_k}(\cdot) \rightarrow F(\cdot)$$

现证 $F_n(\cdot) \xrightarrow{w} F$. 否则 $\exists x_0 \in C(F)$ st $F_n(x_0) \not\xrightarrow{w} F(x_0)$.

又 $\{F_n(x_0)\}$ 为有界列, 必有 $\{F_{m_{n_k}}(x_0)\} \subset \{F_n(x_0)\}$ st $F_{m_{n_k}}(x_0) \rightarrow F^*(x_0) \neq F(x_0)$

对 $\{F_{m_{n_k}}(\cdot)\}$ 重复 step one 证明: $\exists \{F_{m_{n_k}}(\cdot)\}$ st $F_{m_{n_k}}(\cdot) \xrightarrow{w} F(\cdot)$

又 $x_0 \in C(F) \Rightarrow F_{m_{n_k}}(x_0) \rightarrow F(x_0)$ 但同时 $\{F_{m_{n_k}}(x_0)\} \subset \{F_{m_{n_k}}(x_0)\}$

$$\Rightarrow F_{m_{n_k}}(x_0) \rightarrow F^*(x_0) \Rightarrow F^*(x_0) = F(x_0) \text{ 与 } F^*(x_0) \neq F(x_0)$$

矛盾. $\therefore F_n(\cdot) \xrightarrow{w} F(\cdot)$.

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2 题答案:

Proof. (1) $X = z_1 - 2z_2 + z_3$
 $Y = z_1 + 3z_2 - z_3$ 则 $\begin{pmatrix} X \\ Y \end{pmatrix} = \begin{pmatrix} 1 & -2 & 1 \\ 1 & 3 & -1 \end{pmatrix} \begin{pmatrix} z_1 \\ z_2 \\ z_3 \end{pmatrix} \triangleq A \begin{pmatrix} z_1 \\ z_2 \\ z_3 \end{pmatrix}$

则 $(X, Y)^T \sim N\left(A \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}, A \begin{pmatrix} 2 & 2 & 2 \\ 2 & 6 & 8 \\ 2 & 8 & 11 \end{pmatrix} A^T\right)$

$= N\left(\begin{pmatrix} 0 \\ 4 \end{pmatrix}, \begin{pmatrix} 1 & -3 \\ -3 & 27 \end{pmatrix}\right)$

$E(X|Y) = a_1 + r \frac{\sigma_1}{\sigma_2} (Y - a_2)$, where $a_1 = 0, a_2 = 4$
 $= a_1 + \frac{r\sigma_1\sigma_2}{\sigma_2^2} (Y - a_2)$ $\left\{ \begin{array}{l} \sigma_1^2 = 1, \sigma_2^2 = 27, r\sigma_1\sigma_2 = -3 \end{array} \right.$

$= 0 + \frac{-3}{27} (Y - 4) = -\frac{1}{9} (Y - 4) = \frac{4}{9} - \frac{1}{9} (z_1 + 3z_2 - z_3)$

$\therefore E(z_1 - 2z_2 + z_3 | z_1 + 3z_2 - z_3) = \frac{4}{9} - \frac{1}{9} (z_1 + 3z_2 - z_3). \quad \#$

Proof of (2) (1) $\because E z^2 = D(z) = 1, E z = 0$
 $E z^2 z = 0$

$\therefore E(z^2 z) = E z^2 \cdot E z$

则 z^2 与 z 不相关.

(b) $\because P\{|z| < 3\} = 0.999974$ 且概率分布对称则 $\forall c_0 > 1$

又 $\{z^2 < c\} \subset \{|z| < \sqrt{c}\} \subset \{|z| < c\} \subset \{z < c\}$

则 $\exists c > 1$, s.t. $0 < P\{z^2 < c\} < 1$ 且 $0 < P\{z < c\} < 1$

则 $P\{z^2 < c\} P\{z < c\} \neq P\{z < c\} = P\{z^2 < c, z < c\}$.

i.e., z^2 与 z 不相互独立.

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3 题参考答案:

Proof of (1): 利用全概率公式:

$$P\{z_1 < z_2 < z_3\} = \int_{\mathbb{R}} P\{z_1 < z_2 < z_3 \mid z_3 = z\} dF_{z_3}(z)$$

$$= \frac{1}{2} \int_0^2 P\{z_1 < z_2 < z\} dz = \frac{1}{2} \int_0^1 P\{z_1 < z_2 < z\} dz + \frac{1}{2} \int_1^2 P\{z_1 < z_2 < z\} dz$$

$$= \frac{1}{2} \int_0^1 (z + \frac{1}{2}z^2) dz + \frac{1}{2} \int_1^2 (1 + \frac{1}{2}) dz = \frac{13}{24}$$

Proof of (2) $E\{Y^2 \mid z_1\} = E\{(z_1 + z_2)(z_1 + z_3) \mid z_1\}$

$$= E\{z_1^2 + z_1(z_2 + z_3) + z_2z_3 \mid z_1\}$$

$$= E\{z_1^2 \mid z_1\} + E\{(z_2 + z_3)z_1 \mid z_1\} + E\{z_2z_3 \mid z_1\}$$

$$= z_1^2 + z_1 E\{z_2 + z_3\} + E\{z_2z_3\}$$

$$= z_1^2 + z_1 (Ez_2 + Ez_3) + Ez_2 Ez_3$$

$$= z_1^2 + z_1 (\frac{1}{2} + 1) + \frac{1}{2} \times 1 = \frac{1}{2} + \frac{3}{2}z_1 + z_1^2$$

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4 题参考答案:

Proof of (1): $\mathbf{z} \triangleq (z_1, \dots, z_n)^T \sim N(\bar{\mathbf{a}}, B)$, $\bar{\mathbf{a}} = (a_1, \dots, a_n)^T$, $B = \sigma^2 I_{n \times n}$

取正交矩阵 $C = \begin{pmatrix} \frac{1}{\sqrt{n}} & \dots & \frac{1}{\sqrt{n}} \\ * & \dots & * \\ * & \dots & * \end{pmatrix} \triangleq (C_{kj})_{n \times n}$

令 $\vec{y} = C\vec{z}$

$$\text{则 } \vec{\eta} \sim N(C\vec{a}, CBC^T) = N(C\vec{a}, \sigma^2 I_{n \times n})$$

$$\Rightarrow \eta_1, \dots, \eta_n \text{ 相互独立, } \eta_1 = \sqrt{n}\bar{z}, \quad C\vec{a} = \begin{pmatrix} \sqrt{n}a \\ 0 \\ \vdots \\ 0 \end{pmatrix}$$

$$\therefore \eta_1 \sim N(\sqrt{n}a, \sigma^2)$$

$$\eta_k = C_{k1}z_1 + \dots + C_{kn}z_n \sim N(0, \sigma^2) \quad (k=2, \dots, n)$$

$$\eta_1 \text{ 与 } \sum_{k=2}^n \eta_k^2 = \sum_{k=2}^n \eta_k^2 - |\eta_1|^2 = |\vec{z}|^2 - \frac{1}{n} \left(\sum_{k=1}^n z_k \right)^2 = \sum_{k=1}^n (z_k - \bar{z})^2 \text{ 独立}$$

$$\text{i.e., } \bar{z} \text{ 与 } \frac{1}{n-1} \sum_{k=1}^n (z_k - \bar{z})^2$$

$$\text{则 } \frac{(n-1)S_n^2}{\sigma^2} = \frac{1}{\sigma^2} \sum_{k=2}^n \eta_k^2 = \sum_{k=2}^n \left(\frac{\eta_k}{\sigma} \right)^2, \quad \text{又 } \frac{\eta_k}{\sigma} \sim N(0, 1) \text{ 且 } \{\eta_k\} \text{ 独立}$$

$$\text{for } k=2, \text{ 则 } \frac{(n-1)S_n^2}{\sigma^2} \sim \chi_{n-1}^2. \quad \#$$

$$\text{Proof of (2). } \int_0^\infty x^s dF_3(x) = \lim_{M \rightarrow \infty} \int_0^M x^s dF_3(x)$$

$$\int_0^M x^s dF_3(x) = - \int_0^M x^s d(1 - F_3(x)) = -x^s(1 - F_3(x)) \Big|_0^M + \int_0^M s x^{s-1} (1 - F_3(x)) dx$$

$$= -x^s P\{Z \geq x\} \Big|_0^M + \int_0^M s x^{s-1} P\{Z \geq x\} dx$$

$$= -M^s P\{Z \geq M\} + \int_0^M s x^{s-1} P\{Z \geq x\} dx.$$

$$\therefore M^s P\{Z \geq M\} \leq \int_M^\infty x^s dF_3(x) \rightarrow 0 \text{ as } M \rightarrow \infty \quad (\because \int_0^\infty x^s dF_3(x) < \infty)$$

$$\therefore \int_0^\infty x^s dF_3(x) = \lim_{M \rightarrow \infty} \int_0^M s x^{s-1} P\{Z \geq x\} dx$$

$$= \int_0^\infty s x^{s-1} P\{Z \geq x\} dx.$$

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5 题参考答案:

Proof of (1): 令 $\eta_n \triangleq \sum_{k=1}^n I_{\{|Z_k| \leq n\}}$, $S_n^* = \sum_{k=1}^n \eta_k$.

$$\Rightarrow \forall \varepsilon > 0, \sum_{k=1}^{\infty} P\{|Z_k - \eta_k| > \varepsilon\} = \sum_{k=1}^{\infty} P\{Z_k \neq \eta_k\}$$

$$= \sum_{k=1}^{\infty} P\{|Z_k| > k\} = \sum_{k=1}^{\infty} P\{|Z_1| > k\} \leq \int_0^{\infty} |x| dF_{Z_1}(x) = E|Z_1| < +\infty$$

By B-C' lemma $\sum_{k=1}^n \eta_k \xrightarrow{a.s.} 0$, using Toeplitz's lemma,

$$(a) \quad \frac{1}{n} \sum_{k=1}^n (Z_k - \eta_k) = \frac{1}{n} (S_n - S_n^*) \xrightarrow{a.s.} 0$$

$$\& \left(\sum_{n=1}^{\infty} D\left(\frac{\eta_n - E\eta_n}{n}\right) \right) = \sum_{n=1}^{\infty} \frac{D(\eta_n)}{n^2} \leq \sum_{n=1}^{\infty} \frac{E(\eta_n^2)}{n^2}$$

$$= \sum_{n=1}^{\infty} \frac{1}{n^2} \int_{|x| \leq n} x^2 dF_{Z_1}(x) = \sum_{n=1}^{\infty} \frac{1}{n^2} \sum_{j=1}^n \int_{j-1 < |x| \leq j} x^2 dF_{Z_1}(x)$$

$$= \sum_{j=1}^{\infty} \int_{j-1 < |x| \leq j} x^2 dF_{Z_1}(x) \left(\sum_{n=j}^{\infty} \frac{1}{n^2} \right) \quad \left(\because \sum_{n=j}^{\infty} \frac{1}{n^2} \leq \frac{C}{j} \right)$$

$$\leq \sum_{j=1}^{\infty} \frac{C}{j} \int_{j-1 < |x| \leq j} x^2 dF_{Z_1}(x) \leq C \sum_{j=1}^{\infty} \int_{j-1 < |x| \leq j} |x| dF_{Z_1}(x) \leq C E|Z_1| < +\infty$$

$$\Rightarrow \sum_{n=1}^{\infty} \frac{\eta_n - E\eta_n}{n} \text{ converges a.s.},$$

Based on Kronecker's lemma,

$$(b) \quad \frac{1}{n} \sum_{k=1}^n (\eta_k - E\eta_k) \xrightarrow{a.s.} 0$$

$$(c) \quad \& E\eta_n = \int_{\mathbb{R}} x I_{[-n, n]}(x) dF(x). \quad \& |x I_{[-n, n]}(x)| \leq |x| \text{ 且 } x I_{[-n, n]} \xrightarrow{p} 0$$

By Dominated convergence Thm.

$$E\eta_n \rightarrow E Z_1 = a.$$

By Toeplitz' Lemma

$$ES_n^* = \sum_{k=1}^n EY_k.$$

$$\frac{1}{n} \sum_{k=1}^n EY_k = \frac{1}{n} ES_n^* \rightarrow a.$$

利用 (a), (b), (c) $\Rightarrow \frac{1}{n} S_n = \frac{1}{n} (S_n - S_n^*) + \frac{1}{n} (S_n^* - ES_n^*)$
 $+ \frac{1}{n} ES_n^* \xrightarrow{a.s.} a.$

Proof of (2): Assume that $E|Z_1| = +\infty$. #

利用 $\sum_{n=1}^{\infty} P\{|Z_n| \geq n\} \leq E|Z_1| \leq 1 + \sum_{n=1}^{\infty} P\{|Z_n| \geq n\} \quad \forall Z$

$$\Rightarrow E\left\{\frac{|Z_1|}{A}\right\} = +\infty \quad \forall A > 0$$

$$\Rightarrow \sum_{n=1}^{\infty} P\{|Z_n| > An\} = +\infty$$

$$= \left(\sum_{n=1}^{\infty} P\{|Z_n| > An\} \right)$$

$$\Leftrightarrow P\left\{\lim_n \{w: |Z_n| > An\}\right\} = 1$$

$$\left(\begin{aligned} & \text{X: } An \leq |Z_n| \leq |S_n - S_{n-1}| \leq |S_n| + |S_{n-1}| \\ & \therefore |S_n| \geq \frac{A}{2}n \text{ or } |S_{n-1}| > \frac{1}{2}An \end{aligned} \right)$$

$$\Rightarrow P\left\{\lim_n (|S_n| > \frac{A}{2}n)\right\} = 1.$$

$$= P\left\{\bigcap_{k=1}^{\infty} \bigcup_{n=k}^{\infty} \{ |S_n| > \frac{A}{2}n \} \right\} = 1$$

$\bigcap_{A>0} \hat{\Omega}_A$

$$\hat{\Omega} \triangleq \bigcap_{A=1}^{\infty} \hat{\Omega}_A. \Rightarrow P(\hat{\Omega}) = 1, \text{ 且 } \forall w \in \hat{\Omega} \Rightarrow \forall A > 0$$

$$\exists k_0 > 1, \text{ s.t. } \forall w \in \hat{\Omega}, \forall A > 0, w \in \hat{\Omega}_A$$

$$\forall k \geq 1, \exists n \geq k \text{ s.t. } |S_n| > \frac{1}{2}nA$$

$$\Rightarrow \lim_{n \rightarrow \infty} \frac{|S_n(\omega)|}{n} > \frac{1}{2}A, \quad \omega \in \tilde{\Omega}$$

letting $A \rightarrow \infty \Rightarrow$

$$\lim_{n \rightarrow \infty} \frac{|S_n|}{n} = +\infty, \quad \forall \omega \in \tilde{\Omega} \text{ and } P(\tilde{\Omega}) = 1.$$

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6题参考答案:

Proof: ① 设 $T \leq n \Rightarrow S \leq n \Rightarrow |X_T| = |X_T| \sum_{j=1}^n I_{\{T \geq j\}} = \sum_{j=1}^n |X_j| I_{\{T \geq j\}}$

$$\Rightarrow E|Z_T| \leq \sum_{j=1}^n E|X_j| < \infty \text{ similarly, } E|Z_S| \leq \sum_{j=1}^n E|X_j| < \infty$$

i.e., Z_S 和 Z_T 可积

(2) 又 Z_T 和 Z_S 为 $\mathcal{F}_T$ 和 $\mathcal{F}_S$ 可测. $\Rightarrow E\{Z_T | \mathcal{F}_S\} \leq Z_S$

$$\Leftrightarrow \forall A \in \mathcal{F}_S \quad \int_A Z_T dP \leq \int_A Z_S dP.$$

先设 $0 < T - S \leq 1 \Rightarrow A \in \mathcal{F}_S, \quad A = \sum_{j=1}^{n-1} A \cap \{S=j\} \cap \{T=j+1\}$

$$\Rightarrow \int_A Z_T dP = \sum_{j=1}^{n-1} \int_{A \cap \{S=j\} \cap \{T=j+1\}} Z_{j+1} dP \leq \sum_{j=1}^{n-1} \int_{A \cap \{S=j\} \cap \{T=j+1\}} Z_j dP$$

$$= \int_A Z_S dP. \quad (A \cap \{S=j\} \cap \{T=j+1\} \in \mathcal{F}_j)$$

其次 $\Delta R_j = T \wedge (S+j), \quad 1 \leq j \leq n \Rightarrow R_j$ is a stopping time.

且: $S \leq R_1 \leq R_2 \leq \dots \leq R_n = T, \quad R_1 - S \leq 1, \quad R_{j+1} - R_j \leq 1.$

$$\Rightarrow \int_A Z_T dP = \int_A Z_{R_n} dP \leq \int_A Z_{R_{n-1}} dP \leq \dots \leq \int_A Z_S dP.$$

$$\Rightarrow \int_A Z_T dP \leq \int_A Z_S dP \quad \forall A \in \mathcal{F}_S, \quad S \leq T.$$

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7 题答案:

Proof of (1): (1) $\Leftrightarrow \forall A \in \mathcal{G}_1, \int_A E\{Z | \mathcal{G}_2 | \mathcal{G}_1\} d\mathbb{P} = \int_A E\{Z | \mathcal{G}_1\} d\mathbb{P}$

$$\textcircled{*} \text{ 的左边} = \int_A E\{Z | \mathcal{G}_2\} d\mathbb{P} = \int_A Z d\mathbb{P} = \int_A E\{Z | \mathcal{G}_1\} d\mathbb{P} \quad \textcircled{*}$$

($\because A \in \mathcal{G}_2$).

i.e., $\textcircled{*}$ holds.

$$\begin{aligned} \text{Proof of (2): Indeed, } D(Z | \mathcal{G}) &= E\{(Z - E(Z | \mathcal{G}))^2 | \mathcal{G}\} \\ &= E\{Z^2 | \mathcal{G}\} - (E\{Z | \mathcal{G}\})^2 \end{aligned}$$

$$\therefore E\{D(Z | \mathcal{G})\} = E\{Z^2\} - E\{(E(Z | \mathcal{G}))^2\} \quad \textcircled{**}$$

$$\begin{aligned} \text{又 } D(E(Z | \mathcal{G})) &= E\{(E(Z | \mathcal{G}))^2\} - (E(E(Z | \mathcal{G})))^2 \\ &= E\{(E(Z | \mathcal{G}))^2\} - (E(Z))^2 \quad \textcircled{***} \end{aligned}$$

$\textcircled{**} + \textcircled{***}$ 两式相加便得.

$$E\{D(Z | \mathcal{G})\} + D(E(Z | \mathcal{G})) = E\{Z^2\} - (E(Z))^2 = D(Z).$$

i.e., $\textcircled{2}$ holds.

Proof of (3): (a) $f(x) = I_B(x)$, $B \in \mathcal{B}(\mathbb{R}) \Rightarrow$

$$\int_{\mathbb{R}} I_B(x) d\mathbb{P} = \mathbb{P}\{x: x \in B\} = \mathbb{P}_Z(B) = \int_{\mathbb{R}} I_B(x) d\mathbb{F}_Z(x)$$

$$\textcircled{b} \quad \forall f \geq 0 \quad \exists \quad f_n(x) = \sum_{k=0}^{n2^n-1} \frac{k}{2^n} I_{\{x: \frac{k}{2^n} \leq f(x) < \frac{k+1}{2^n}\}} = \sum_{k=0}^{n2^n-1} \frac{k}{2^n} I_{B_k^{(n)}}$$

$$B_k \triangleq \{x: \frac{k}{2^n} \leq f(x) < \frac{k+1}{2^n}\}, \quad B_{n2^n} = \{x: f(x) \geq n\}$$

则 $B_k \in \beta(\mathbb{R})$, $k=0, \dots, k=R^n-1$, 且 $f_n(x) \uparrow f(x)$.

$$\text{By (a), } \int_{\Omega} f_n(\xi, \omega) d\mathbb{P} = \int_{\mathbb{R}} f_n(x) d\mathbb{F}_\xi(x).$$

在上式令 $n \rightarrow \infty$, 由单调收敛定理, 有

$$\int_{\Omega} f(\xi, \omega) d\mathbb{P} = \int_{\mathbb{R}} f(x) d\mathbb{F}_\xi(x).$$

$$(c) f = f^+ - f^-, \because E f^\pm(\xi) \leq E |f(\xi)| < \infty.$$

$$\text{则 by (b) } \int_{\Omega} f^\pm(\xi, \omega) d\mathbb{P} = \int_{\mathbb{R}} f^\pm(x) d\mathbb{F}_\xi(x). \Rightarrow$$

$$\int_{\Omega} f(\xi, \omega) d\mathbb{P} = \int_{\mathbb{R}} f(x) d\mathbb{F}_\xi(x).$$

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8题答案:

$$\text{Proof of (1): } E \xi_1^2 = \int_0^\infty 2x P\{|\xi_1| \geq x\} dx \geq \int_0^\infty 2x \cdot \frac{1}{x^2} dx = \infty$$

$$\text{Proof of (2): } S_n = \sum_{k=1}^n \xi_k, \quad Y_{n,m} \triangleq \xi_m I_{\{|\xi_m| \leq n^{\frac{1}{2}} \log \log n\}}$$

$$\text{则 } \because P\{\xi_1 \geq x\} = P\{\xi_1 \leq -x\} \Rightarrow E Y_{n,m} = 0$$

$$\text{令 } S_n^* = \sum_{k=1}^n Y_{n,k}, \text{ 则}$$

$$c_n = n^{\frac{1}{2}} \log \log n$$

$$P\{S_n \neq S_n^*\} \leq \sum_{m=1}^n P\{Y_{n,m} \neq \xi_m\} = \sum_{m=1}^n P\{|\xi_m| > c_n\}$$

$$= \sum_{m=1}^n P\{|\xi_1| > c_n\} = n P\{|\xi_1| > c_n\} = n \cdot \frac{1}{c_n^2} \rightarrow 0$$

$$\Rightarrow S_n - S_n^* \xrightarrow{P} 0$$

$$\text{下证 } \textcircled{a} \left(\frac{\sum_{m=1}^n Y_{n,m}}{\sqrt{n \log n}} \right) \text{ 令 } \tilde{\xi}_{n,m} \triangleq \frac{Y_{n,m}}{\sqrt{n \log n}} \text{ 则 } E \tilde{\xi}_{n,m} = 0;$$

$$\textcircled{b} \sum_{k=1}^n E \xi_{n,k}^2 \rightarrow 1; \quad \textcircled{c} \forall \varepsilon > 0, \lim_{n \rightarrow \infty} \sum_{m=1}^n E(|\tilde{\xi}_{n,m}|^2 I_{\{|\tilde{\xi}_{n,m}| > \varepsilon\}}) = 0$$

(a) 是显然的.

$$(b) \sum_{k=1}^n E Z_{n,k}^2 = \sum_{k=1}^n \frac{E Y_{n,m}^2}{n \log n} = \frac{1}{n \log n} \sum_{k=1}^n E Y_{n,k}^2.$$

$$\therefore P\{|Y_{n,m}| \geq x\} \leq P\{|Z_1| \geq x\} \text{ and } P\{|Y_{n,m}| \geq x\} = 0 \text{ for } x > c_n$$

$$\therefore E Y_{n,m}^2 \leq \int_0^{c_n} 2y P\{|Z_1| > y\} dy = 1 + \int_1^{c_n} \frac{2}{y} dy$$

$$= 1 + 2 \log c_n = 1 + \log n + 2 \log \log \log n \sim \log n \quad (n \rightarrow \infty)$$

$$\text{同理 } P\{|Y_{n,m}| \geq x\} = P\{|Z_1| \geq x\} - P\{|Z_1| \geq c_n\}$$

$$\text{注意到: } P\{|Z_1| \geq c_n\} = \frac{1}{n (\log \log n)^2} \leq \frac{1}{x^2 (\log \log n)^2} \quad x \leq \sqrt{n}$$

$$= \frac{1}{(\log \log n)^2} P\{|Z_1| > x\} \quad x >$$

$$\text{we have } P\{|Y_{n,m}| \geq x\} \geq (1 - (\log \log n)^{-2}) P\{|Z_1| > x\} \text{ for } x \leq \sqrt{n},$$

$$\text{因此 } E Y_{n,m}^2 = \int_0^{c_n} 2x P\{|Y_{n,m}| \geq x\} dx$$

$$\geq 2 \int_1^{\sqrt{n}} x (1 - (\log \log n)^{-2}) P\{|Z_1| > x\} dx$$

$$= (1 - (\log \log n)^{-2}) 2 \int_1^{\sqrt{n}} \frac{1}{x} dx = [1 - (\log \log n)^{-2}] \log n$$

$$\sim \log n$$

$$\therefore E Y_{n,m}^2 \sim \log n \quad (n \rightarrow \infty)$$

$$\therefore \sum_{k=1}^n E Z_{n,k}^2 = \frac{1}{n} \sum_{k=1}^n \frac{E Y_{n,m}^2}{\log n} \rightarrow 1 \quad (n \rightarrow \infty)$$

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④

$$(c) \because |Z_{n,m}| \leq \frac{\log \log n}{(\log n)^{\frac{1}{2}}} \rightarrow 0, \text{ for large } n, \text{ If } |Z_{n,m}| \geq \delta$$

Using (*)

Thus, $\forall \varepsilon > 0$,

$$\lim_{n \rightarrow \infty} \sum_{m=1}^n E\{|Z_{n,m}|^2 I_{\{|Z_{n,m}| > \varepsilon\}}\} = 0$$

i.e., (c) holds.

最后证明 $S_n^{**} = \sum_{k=1}^n Z_{n,k} = \frac{S_n^*}{\sqrt{n \log n}} \xrightarrow{W} Z \sim N(0,1)$

$$f_n(t) \triangleq E\{e^{it S_n^{**}}\} = \prod_{k=1}^n f_{n,k}(t) \quad f_{n,k}(t) = E\{e^{it Z_{n,k}}\}$$

要证明: $f_n(t) \rightarrow e^{-\frac{1}{2}t^2}$

$$\sigma_{n,k}^2 = E Z_{n,k}^2$$

$$|f_n(t) - e^{-\frac{1}{2}t^2}| = \left| \prod_{k=1}^n f_{n,k}(t) - e^{-\frac{1}{2}t^2} \right| \quad f_{n,k}(t) = 1 - \frac{1}{2} \sigma_{n,k}^2 t^2 + o(\sigma_{n,k}^2)$$

$$\leq \left| \prod_{k=1}^n f_{n,k}(t) - \prod_{k=1}^n \left(1 - \frac{1}{2} \sigma_{n,k}^2 t^2\right) \right| + \left| \prod_{k=1}^n \left(1 - \frac{1}{2} \sigma_{n,k}^2 t^2\right) - e^{-\frac{1}{2}t^2} \right|$$

$$\triangleq \Omega_1(n) + \Omega_2(n)$$

①

$$\Omega_1(n) \leq \sum_{k=1}^n |f_{n,k}(t) - (1 - \frac{1}{2} \sigma_{n,k}^2 t^2)|$$

$$= \sum_{k=1}^n |E\{e^{it Z_{n,k}} - 1 + it Z_{n,k} + \frac{1}{2} (it Z_{n,k})^2\}|$$

$$\leq \sum_{k=1}^n E\{\min\{|t Z_{n,k}|^3, |t Z_{n,k}|^2\}\}$$

$$= \sum_{k=1}^n [E\{|t Z_{n,k}|^3 I_{\{|Z_{n,k}| < \varepsilon\}} + |t Z_{n,k}|^2 I_{\{|Z_{n,k}| \geq \varepsilon\}}\}]$$

$$\leq |t|^3 \varepsilon \sum_{k=1}^n \sigma_{n,k}^2 + t^2 \sum_{k=1}^n E\{Z_{n,k}^2 I_{\{|Z_{n,k}| \geq \varepsilon\}}\}$$

$$\rightarrow 0 \quad (\text{as } n \rightarrow \infty \text{ and then } \varepsilon \rightarrow 0) \quad \textcircled{2}$$

$$\Omega_2(n) \leq \left| \prod_{k=1}^n \left(1 - \frac{1}{2} \sigma_{n,k}^2 t^2\right) - \prod_{k=1}^n e^{-\frac{1}{2} \sigma_{n,k}^2 t^2} \right| + \left| \prod_{k=1}^n e^{-\frac{1}{2} \sigma_{n,k}^2 t^2} - e^{-\frac{1}{2} t^2} \right|$$

$$= \Omega_{21}(n) + \Omega_{22}(n)$$

$$\begin{aligned}
 |\square_{21}(n)| &\leq \sum_{k=1}^n \left| e^{-\frac{1}{2}\sigma_{n,k}^2 t^2} - \left(1 - \frac{1}{2}\sigma_{n,k}^2 t^2\right) \right| \\
 &\leq \sum_{k=1}^n \left| -\frac{1}{2}\sigma_{n,k}^2 t^2 \right| \leq \frac{1}{4} t^2 \max_{1 \leq k \leq n} \{\sigma_{n,k}^2\} \sum_{k=1}^n \sigma_{n,k}^2 \\
 &\leq \frac{1}{4} t^2 \max_{1 \leq k \leq n} \{\sigma_{n,k}^2\} \quad (3)
 \end{aligned}$$

$$\begin{aligned}
 \text{又 } \sigma_{n,k}^2 &= E\beta_{n,k}^2 \leq \varepsilon^2 + E\{\beta_{n,k}^2 I_{\{|\beta_{n,k}| \geq \varepsilon\}}\} \\
 &\leq \varepsilon^2 + \sum_{k=1}^n E\{\beta_{n,k}^2 I_{\{|\beta_{n,k}| \geq \varepsilon\}}\} \rightarrow 0 \quad (n \rightarrow \infty, \varepsilon \rightarrow 0)
 \end{aligned}$$

$$\therefore |\square_{21}(n)| \rightarrow 0 \quad (4)$$

$$\text{又 } |\square_{22}(n)| = \left| e^{-\frac{1}{2} \sum_{k=1}^n \sigma_{n,k}^2 t^2} - e^{-\frac{1}{2} t^2} \right| \rightarrow 0 \quad \left(\text{as } \sum_{k=1}^n \sigma_{n,k}^2 = \sum_{k=1}^n E\{\beta_{n,k}^2\} \rightarrow 1 \right)$$

$$\text{利用 (1), (2), (3) and (4)} \Rightarrow f_n(t) \rightarrow e^{-\frac{1}{2} t^2}.$$

$$\text{证 } \frac{S_n^*}{\sqrt{n \log n}} \xrightarrow{d} Z \sim N(0,1) \quad (5)$$

$$\text{又 } S_n - S_n^* \xrightarrow{P} 0 \Rightarrow \frac{1}{\sqrt{n \log n}} (S_n - S_n^*) \xrightarrow{P} 0$$

$$\therefore \frac{1}{\sqrt{n \log n}} (S_n - S_n^*) \xrightarrow{d} 0 \quad (6)$$

$$\begin{aligned}
 \text{利用 (5) 和 (6) 可知: } \frac{S_n}{\sqrt{n \log n}} &= \frac{1}{\sqrt{n \log n}} (S_n - S_n^*) + \frac{1}{\sqrt{n \log n}} S_n^* \\
 &\xrightarrow{d} Z \sim N(0,1).
 \end{aligned}$$

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